Zero-Shot Topic Labeling: Extensions and Model Comparison

Supervisor Meeting Summary

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September 3, 2026

1 Starting Point

- Base paper (BTW2025-122): zero-shot topic labeling with Llama 3 8B, 19-topic vocabulary, 2500 papers.
- Reported baseline: 50% success, hallucination 25% (titles) / 33% (abstracts).
- Two extensions investigated: (1) fixing the matching step, (2) testing newer models and model size.

2 Extension 1: Matching Fix

- Original matching: exact string match only, breaks on quotes or near-miss wording.
- Fix: strip punctuation, then word-containment fuzzy match, with the ontology's parent category (`computer science`) explicitly excluded from auto-resolution.

Input	Old matching	New matching
`"'software engineering'"`	hallucination	software engineering
`computer vision`	hallucination	computer imaging and vision
`computer science`	hallucination	hallucination (correct)

At full scale (Qwen3-8B, titles): 103/2500 answers (4.1%) recovered by the fix, all the literal string `computer vision`, 83 became successes, 20 became misclassified, 0 stayed unresolved.

3 Extension 2: Newer Models vs. Paper Baseline

Model	Success	Misclassified	Hallucination
Paper (Llama 3 8B), titles	50%	25%	25%
Paper (Llama 3 8B), abstracts	50%	17%	33%
Qwen3-8B, titles	68.2%	29.3%	2.5%
Qwen3-8B, abstracts	70.0%	27.2%	2.8%
Claude Haiku 4.5, titles	70.9%	27.6%	1.5%
Claude Haiku 4.5, abstracts	74.0%	23.3%	2.7%

- Every newer model beats the paper by a wide margin, driven mainly by hallucination collapsing to under 4%.
- Titles vs. abstracts: at native precision, Qwen3-8B and Claude both favor abstracts, matching every Qwen3.5 size tested in Section 4. An earlier GGUF-quantized run of Qwen3-8B had shown the opposite (titles ahead), which in hindsight looks like a precision artifact of that specific backend, not a genuine per-model difference.

4 Extension 3: Does Model Size Matter?

Same task, same prompt, Qwen3.5 family at five sizes (0.8B to 27B).

Size	Titles	Abstracts
0.8B	52.1%	53.7%
4B	75.4%	76.7%
9B	78.6%	80.8%
27B	76.0%	77.0%

- **9B achieves the highest accuracy**, not the largest model tested. Scaling helps sharply to 4B, a little more to 9B, then reverses at 27B.
- 0.8B already beats the paper's Llama 3 8B baseline (52.1% vs 50%), a model ten times its size, from an older generation.

5 Bonus Finding: Quantization Sensitivity Scales With Size

- **What:** storing and computing a model's weights with fewer bits than the precision it was trained in. Here: 16-bit floating point (bf16) reduced to 4-bit. A lossy compression of the model's parameters, not the model itself.
- **How:** weights are grouped into small blocks, each block gets a scale factor so its original range of values can be represented with only 16 discrete levels (4 bits) instead of the full 16-bit range. Weights are converted back to a working precision ("dequantized") just before each computation, so this introduces a small, bounded amount of numerical error per weight, not a change to the model's architecture or training.
- **Why:** roughly 4x less memory per parameter. This is what let the 27B checkpoint (needs $\sim$54GB in bf16) fit entirely on a single 24GB GPU, avoiding a much slower fallback where the excess spills into system RAM. It can also increase inference speed, since less data has to move through memory per token generated.

Size	Field	bf16	4-bit	Difference
0.8B	titles	52.1%	39.2%	**-12.9 pts**
0.8B	abstracts	53.7%	27.6%	**-26.1 pts**
4B	titles	75.4%	74.6%	-0.8 pts
9B	titles	78.6%	80.0%	+1.4 pts
27B	titles	75.4%	76.0%	+0.6 pts

- Small models are fragile under 4-bit quantization, 0.8B loses over a quarter of its accuracy edge.
- 4B and up are effectively indifferent, gains and losses under 1.5 points.
- Found by accident (a script bug applied quantization to every size, not only 27B), but every repeat run came back bit-identical, generation is fully deterministic, so this is a real precision effect, not noise.

Run	Success	Misclassified	Hallucination
Paper (Llama 3 8B), titles	50%	25%	25%
Paper (Llama 3 8B), abstracts	50%	17%	33%
Qwen3.5-0.8B, titles	52.1%	47.3%	0.6%
Qwen3.5-0.8B, abstracts	53.7%	43.9%	2.4%
Qwen3-8B, titles	68.2%	29.3%	2.5%
Qwen3-8B, abstracts	70.0%	27.2%	2.8%
Claude Haiku 4.5, titles	70.9%	27.6%	1.5%
Claude Haiku 4.5, abstracts	74.0%	23.3%	2.7%
Qwen3.5-4B, titles	75.4%	24.0%	0.6%
Qwen3.5-4B, abstracts	76.7%	22.3%	1.0%
Qwen3.5-9B, titles	78.6%	20.8%	0.6%
Qwen3.5-9B, abstracts	80.8%	18.7%	0.5%
Qwen3.5-27B, titles (4-bit)	76.0%	24.0%	0.04%
Qwen3.5-27B, abstracts (4-bit)	77.0%	22.9%	0.1%

6 Full Results Table

7 Conclusion

- **Extension 1 (matching fix):** mainly moves the hallucination column, not the model's actual answers. 4.1% of Qwen3-8B title answers were wrongly counted as hallucination due to fragile exact-string matching; the fix recovered most of them into success. Part of the paper's reported hallucination rate is a measurement artifact, not only a model failure.

- **Extension 2 (newer models):** mainly collapses hallucination, from 25–33% (paper) to under 4% for every newer model tested, while success rises 17–30 points and misclassified stays roughly flat or rises slightly. Newer, better instruction-tuned models are far better at staying inside the controlled vocabulary; that alone accounts for most of the improvement over the paper's baseline, not necessarily better topic judgment.

- **Extension 3 (model size):** hallucination stays low and flat across every size tested (0.6% to 2.4%, no trend with size), so hallucination is a generation/training effect, not a size effect. What moves with size is the success/misclassified split: 0.8B misclassifies nearly half the time (47.3%/43.9%), 9B misclassifies about a fifth of the time. Size affects whether the model picks the *correct* topic, not whether it stays inside the vocabulary, and this effect does not improve indefinitely, it peaks at 9B and reverses at 27B.

- **Taken together:** the three extensions target different parts of the original failure mode. The matching fix and newer models both attack hallucination, one through evaluation robustness, the other through instruction-following, while model size mainly shifts the success/misclassified balance, that is, topic judgment, with diminishing and eventually reversing returns.

8 Open Questions for Discussion

- Which direction next: cost optimization and multi-label evaluation, or few-shot prompting?
- Worth running Qwen3.5-2B to fill the one gap in the size sweep?
- How to frame the 9B-highest-accuracy finding in the thesis narrative?